Notes: Duchin and Sosyura (JFE, 2012)

The Politics of Government Investment

Contents

1 Big picture

1.1 Question

Duchin and Sosyura study whether corporate political connections affect the allocation and performance of government investment funds. The empirical setting is the **Capital Purchase Program (CPP)**, the first and largest initiative under the **Troubled Asset Relief Program (TARP)** during the 2008–2009 financial crisis.

> The central question is whether federal investment decisions were driven only by bank fundamentals and systemic importance, or whether politically connected financial institutions received preferential access to government capital.

The paper separates three decisions: whether an eligible firm applies for CPP funds, whether regulators approve the application, and how recipients perform after receiving government capital. This separation is important because political influence could matter for demand for capital, for government allocation, or for the quality of funded investments.

1.2 Setting

CPP was announced in October 2008. Instead of purchasing troubled assets directly, the Treasury invested capital in qualifying financial institutions. A participating bank issued preferred stock to the Treasury and, for public firms, warrants on common stock. The program ultimately invested about **$204.9 billion** in **707 financial institutions**.

The paper studies public financial institutions because application status can be inferred from public filings and because stock-market data allow the authors to evaluate investment performance. Public CPP recipients account for most of the program economically: the public recipients in the data obtained **$190.1 billion**, or about **92.7%** of CPP capital invested in public and private firms combined.

1.3 Core challenge

The central empirical challenge is that political connections are not randomly assigned. Large banks, older banks, distressed banks, and systemically important institutions may be more politically connected and also more likely to receive government support. A simple comparison of connected and unconnected firms would therefore mix political influence with bank fundamentals and financial stability concerns.

Interpretation. The paper is not simply asking whether connected banks look different. It asks whether, after controlling for the financial criteria regulators said they used, political connections still predict approval and whether connected recipients subsequently perform worse.

1.4 Hypotheses

The paper frames the evidence around three competing views.

- **Political-influence or agency view.** Connected firms receive favorable treatment because politicians or regulators accommodate private political interests. This predicts higher approval rates for connected firms and lower ex post returns on investments in those firms.
- **Information view.** Connections may transmit useful information from firms to regulators during a crisis. This predicts higher approval rates for connected firms, but the funded connected firms should perform at least as well as unconnected firms.
- **No-effect view.** Public scrutiny, disclosure of lobbying and contributions, and audit risk may neutralize political influence. This predicts no meaningful difference between connected and unconnected firms in approvals or performance.

1.5 Main contribution

- The paper hand-collects application status for CPP-eligible public firms, which is valuable because regulators did not publicly disclose all applications.
- It measures political influence in several distinct ways: electoral district ties to House Financial Services subcommittees, connected directors, lobbying, and campaign contributions.
- It distinguishes the application decision from the approval decision, which helps separate firms demand for aid from regulators allocation of aid.
- It studies ex post performance, thereby distinguishing the information view from the political-distortion view.
- It conducts robustness tests for systemic importance, size, state-level matching, placebo connections, and pre-CPP performance differences.

1.6 Main result

Political connections do **not** reliably predict whether an eligible public firm applies for CPP funds. This is consistent with the fact that applying was cheap, widely understood, and attractive during a capital-market freeze.

Political connections do predict **application approval**. Conditional on applying and controlling for bank fundamentals, industry fixed effects, and regulator fixed effects, the probability of approval is higher for firms with ties to relevant House subcommittees, connected board members, targeted lobbying expenditures, and targeted campaign contributions.

The magnitudes are economically meaningful. The paper reports that firms headquartered in districts of key House Financial Services subcommittee members were about **6.3 percentage points** more likely to be approved; firms employing a connected director were about **9.1 percentage points** more likely to be approved; a one-standard-deviation increase in relevant lobbying was associated with about **7.6 percentage points** higher approval probability; and a one-standard-deviation increase in relevant campaign contributions was associated with about **5.0 percentage points** higher approval probability.

After receiving CPP funds, politically connected firms underperform unconnected recipients. A one-standard-deviation increase in the political connections index is associated with an after-CPP decline of about **0.16–0.17 percentage points in quarterly ROA** and about **1.14–1.20 percentage

points in quarterly market-adjusted returns.

1.7 Economic takeaway

The evidence is most consistent with the political-distortion view. Connections appear to improve access to government capital, but investments in connected firms deliver worse financial outcomes. The results therefore suggest that political influence can distort government investment efficiency even in a setting with public disclosure, formal eligibility criteria, and regulatory oversight.

2 Data and measurement

2.1 CPP institutional background

The Emergency Economic Stabilization Act of October 2008 authorized TARP. CPP was the capital-injection program under which the Treasury invested in qualifying financial institutions. Qualifying firms submitted a short application to their primary banking regulator–the Federal Reserve, FDIC, OCC, or OTS. If the regulator supported the application, the file was forwarded to the Treasury, which made the final investment decision.

CPP investments took the form of cumulative perpetual preferred stock. The dividend rate was **5% annually for the first five years** and **9% thereafter**. For public firms, the Treasury also received warrants. The preferred-stock investment was generally between **1% and 3% of risk-weighted assets**, subject to a maximum of $25 billion.

Interpretation. The application process creates a useful empirical structure: the authors can ask separately whether political connections affect the choice to ask for funds and whether they affect regulators and Treasury approval conditional on asking.

2.2 Sample construction

The initial population contains **600 public domestically controlled financial institutions** eligible for CPP and active as of September 30, 2008. The authors identify application status from quarterly filings, annual reports, proxy statements, press releases, and, when necessary, investor-relations contacts.

- **537 firms** have known application status, representing 89.5% of eligible public firms.
- The paper excludes the first wave of the **eight largest CPP investments** announced at program initiation: Citigroup, JP Morgan, Bank of America including Merrill Lynch, Goldman Sachs, Morgan Stanley, State Street, Bank of New York Mellon, and Wells Fargo including Wachovia.
- The final main sample contains **529 firms**.
- Of these, **424 firms** applied for CPP funds and **105 firms** explicitly did not apply.
- Among the applicants, **337** were approved and **87** were not approved.
- Among approved firms, **286** accepted CPP funds and **51** declined them.

2.3 Political connection measures

The paper uses four primary measures of political influence. Each is designed to capture a different channel through which a firm could affect CPP decisions.

- **House Financial Services Subcommittee connection.** A firm is connected if the House representative for the district of the firm headquarters served on the Capital Markets Subcommittee or the Financial Institutions Subcommittee of the House Financial Services Committee in 2008 or 2009.
- **Connected board member.** A firm is connected if its board in 2008–2009 included a director with current or former experience at the Treasury, Congress, or a banking regulator such as the Federal Reserve, FDIC, OTS, or OCC.
- **Lobbying.** The measure is lobbying expenditure targeted at Congress, Treasury, or banking regulators on banking, financial-institution, or bankruptcy issues from 2008:Q1 to 2009:Q1, scaled by book assets.
- **Campaign contributions.** The measure is PAC contributions in the 2008 election cycle to members of the House Financial Services Committee subcommittees most relevant for CPP, scaled by book assets.

2.4 Political connections index

To aggregate the four channels, the authors construct a political connections index. For each firm, they compute the percentile rank for each political measure and average the ranks. The resulting index is normalized between zero and one.

$$\text{Connection Index}_i = \frac{1}{4}\sum_{m=1}^{4} \text{PercentileRank}\left(P_{im}\right).$$

Interpretation. A value of 0.90 means that the firm is, on average, around the ninetieth percentile of political connectedness among eligible firms. The index captures overall political intensity rather than reliance on one single political channel.

2.5 Financial controls

The paper controls for financial condition using proxies for the **CAMELS** dimensions used by bank regulators. The true CAMELS ratings are confidential, so the authors use public and regulatory financial data to construct proxies.

- **Capital adequacy:** Tier 1 risk-based capital ratio.
- **Asset quality:** negative of net loan losses relative to average loans and leases.
- **Management quality:** negative of the number of disciplinary or enforcement actions.
- **Earnings:** return on assets.
- **Liquidity:** cash relative to deposits.

- **Sensitivity to market risk:** maturity gap between short-term assets and liabilities relative to earning assets.

Additional controls include size, market capitalization, age, leverage, deposit-to-asset ratio, and foreclosure exposure. The main specifications also include regulator fixed effects and two-digit SIC industry fixed effects.

2.6 Outcome variables

- **CPP application:** equals one if the eligible firm submitted a CPP application.
- **CPP approval:** equals one if the application was approved, conditional on applying.
- **CPP investment:** equals one if the firm accepted CPP funds, conditional on approval.
- **Performance outcomes:** ROA, Tobin's Q, an alternative Tobin's Q, raw returns, market-adjusted returns, and industry-adjusted returns, measured quarterly.

2.7 Descriptive facts

Table 1 reports that 23.4% of firms had a House Financial Services subcommittee connection, 29.9% had at least one connected board member, 6.8% engaged in relevant lobbying, and 5.1% made relevant PAC contributions. The average political connections index is about 0.50, with a standard deviation of 0.108.

The average CPP recipient in the final sample received about **$263million**, whilethemedianrecipientreceivedabo$
$*$32.1 million**. The distribution is therefore highly skewed, as expected in banking.

3 Models and empirical specifications

3.1 Application decision

The first empirical model asks whether political connections predict the decision to apply for CPP funds.

$$\Pr(\text{Apply}_i = 1) = \Lambda\left(\alpha + \beta\,\text{Politics}_i + \Gamma'\text{CAMELS}_i + \Delta' X_i + \mu_{r(i)} + \eta_{s(i)}\right).$$

Interpretation. Here $\Lambda(\cdot)$ is the logit link. $extPolitics_i$ is either one political measure or the aggregate political connections index. $extCAMELS_i$ are financial-condition proxies, X_i are other firm controls, $\mu_{r(i)}$ are regulator fixed effects, and $\eta_{s(i)}$ are industry fixed effects. The coefficient β asks whether connected firms are more likely to ask for CPP funds.

The application model uses the 529 eligible public firms in the main sample. The paper finds that political connections do not have a robust association with application. Financial need matters more: weaker capital, more leverage, greater foreclosure exposure, and larger market-risk exposure are associated with applying.

3.2 Approval decision

The second model asks whether political connections predict approval, conditional on application.

$$\Pr(\text{Approved}_i = 1 \mid \text{Apply}_i = 1) = \Lambda\left(\alpha + \beta\,\text{Politics}_i + \Gamma'\text{CAMELS}_i + \Delta' X_i + \mu_{r(i)} + \eta_{s(i)}\right).$$

Interpretation. This is the central allocation specification. Because the sample is restricted to applicants, β captures differences in regulators and Treasury approval rather than differences in firms willingness to request aid.

The approval model uses the 424 firms that applied for CPP. In contrast to the application model, each measure of political influence is positively associated with approval. Earnings, market capitalization, and age also tend to predict approval, consistent with regulators favoring stronger and more established institutions.

3.3 Marginal-effect interpretation

The logit coefficients are converted into economic magnitudes by computing marginal effects while holding other variables at their means. For continuous political measures such as lobbying and contributions, the paper reports the change in approval probability associated with a move from one-half standard deviation below the mean to one-half standard deviation above the mean.

$$\Delta\Pr(\text{Approved}) = \Pr(\text{Approved} \mid P_i = \bar{P} + 0.5\sigma_P) - \Pr(\text{Approved} \mid P_i = \bar{P} - 0.5\sigma_P).$$

Interpretation. This standardization makes the lobbying and contribution effects comparable to indicator-variable effects such as having a connected director or being located in a connected congressional district.

3.4 Robustness specifications

The robustness section keeps the approval model but changes the sample, controls, or political variables. The main alternatives are:

- alternative CAMELS proxies;
- dropping the top and bottom quartiles of capital adequacy;
- excluding firms headquartered in New York;
- using a dummy rather than an index for House subcommittee ties;
- using unscaled dollar amounts of lobbying and contributions;
- estimating on all 600 qualifying public institutions rather than only the main sample;
- replacing true political measures with placebo political measures.

Interpretation. The core result is robust: true contemporaneous political connections predict approval, while placebo connections generally do not.

3.5 Systemic importance specifications

A major alternative explanation is that connected firms were not politically favored; they were simply more systemically important. The paper therefore estimates variants of the approval model that explicitly address systemic importance.

$$\Pr(\text{Approved}_i = 1) = \Lambda\left(\alpha + \beta\,\text{ConnectionIndex}_i + \theta\,\text{SystemicImportance}_i + \Gamma'\text{Controls}_i + \mu_{r(i)} + \eta_{s(i)}\right).$$

Systemic importance is measured several ways: excluding the largest 5% of firms by size, adding higher-order powers of size, including a Peer Group 1 indicator for banks with at least $3 billion in assets, estimating size-matched and state-size-matched samples, and adding DCoVaR as a market-based measure of systemic risk.

Interpretation. The political connections index remains positive and significant across these tests. This is a key part of the paper because it addresses the concern that politics is only proxying for too-big-to-fail status.

3.6 Investment-performance tests

The performance tests ask whether the connected firms that received government capital performed better or worse than unconnected recipients after CPP. This is the main test distinguishing the information view from the political-distortion view.

$$Y_{it} = \alpha_i + \tau\,\text{AfterCPP}_t + \beta\,(\text{AfterCPP}_t \times \text{ConnectionIndex}_i) + \Gamma' X_{it} + \varepsilon_{it}.$$

Interpretation. The dependent variable Y_{it} is a quarterly performance measure such as ROA, Tobin's Q, or stock returns. Bank fixed effects absorb time-invariant differences across recipients. The coefficient of interest is β. A negative β means that connected recipients performed worse after CPP relative to unconnected recipients.

The authors first show that connected and unconnected recipients did not have statistically different pre-CPP performance trends. Then they show that the interaction of After CPP with the connections index is negative across ROA, Tobin's Q, raw returns, market-adjusted returns, and industry-adjusted returns. The negative effect survives time-varying bank controls.

3.7 Standard errors and fixed effects

The application and approval regressions report heteroskedasticity-consistent standard errors. The performance regressions cluster standard errors at the bank level and include bank fixed effects in the difference-in-differences specifications. Regulator fixed effects are important because different primary regulators may have applied different standards or had different informational advantages.

4 Identification and instruments

4.1 What identifies the estimate?

The paper does not use a formal instrumental-variable design. Identification comes from comparing connected and unconnected eligible public financial institutions within the same national program, after controlling for financial condition, size, age, foreclosure exposure, regulator fixed effects, and industry fixed effects.

The most important design choice is to separate application from approval. If connected firms were simply more aware of CPP or more willing to interact with government, political variables would predict application. They do not. Political variables become important at the approval stage, where regulators and the Treasury decide who receives capital.

4.2 Why reverse causality is less central here

The political variables are measured before or during the administration of CPP. Some are long-standing or predetermined relative to the application decision, such as the congressional district of a firm headquarters or the prior employment history of directors. Lobbying and contributions are more active channels, but the paper targets lobbying and contributions to the agencies and politicians most relevant for CPP.

Interpretation. The results should be interpreted as evidence of a strong association consistent with political influence, not as a randomized causal experiment. The paper is careful to address many alternative explanations, but it cannot make connections randomly assigned.

4.3 Controls for financial condition

Regulators stated that CPP approval depended on the strength and viability of the applicant. The CAMELS-style controls therefore matter. They are meant to capture exactly the financial fundamentals that would rationally determine approval: capital adequacy, asset quality, management quality, earnings, liquidity, and interest-rate risk exposure.

In the approval regressions, higher earnings, larger market capitalization, and older firm age tend to predict approval. This means the model is not mechanically attributing all approval differences to politics; it also recognizes that stronger firms were favored.

4.4 Systemic-importance concern

One major threat is that politically connected firms might be larger or more systemically important. The paper handles this in several ways. It excludes the eight largest initial CPP recipients, adds size controls, excludes the largest 5% of firms, adds nonlinear size terms, includes a Peer Group 1 dummy, uses size-matched and state-size-matched samples, and controls for DCoVaR.

Economic message: This is economically important because the policy objective of preserving financial stability could justify aid to large or interconnected banks. The robustness tests show that political connections remain relevant even after substantial attempts to control for systemic importance.

4.5 Placebo political variables

The placebo tests ask whether any variable that looks like politics predicts approval, or whether only politically relevant contemporaneous connections do. The placebo measures include director connections that existed in 2006–2007 but not during 2008–2009, House committee connections from an earlier Congress but not the CPP period, lobbying to agencies unrelated to CPP, and contributions to losing challengers.

Interpretation. The placebo variables generally do not predict CPP approval. This supports the interpretation that the main effects are tied to channels plausibly relevant for CPP rather than to generic firm traits correlated with political activity.

4.6 Performance evidence as a mechanism test

The performance tests are not only about returns. They also help distinguish two theories of why connected firms were approved. If connections conveyed valuable private information, connected recipients should not underperform after receiving funds. Instead, the connected recipients perform worse. This pushes the interpretation toward political distortion rather than information transmission.

4.7 Remaining limitations

- The evidence is not based on random assignment or a formal instrument.
- The sample is limited to public financial institutions, because application status and performance are observable for these firms.
- Financial returns may not capture all social objectives of CPP, such as financial stability, local lending, or employment preservation.
- Some forms of political influence may be unobserved or informal, and the observed measures may be proxies for broader networks.

Even with these limitations, the combination of approval regressions, robustness tests, placebo tests, and underperformance evidence makes the political-distortion interpretation credible.

5 Tables and figures

5.1 Figure 1: Sample firms and CPP applications

Read: The figure starts with 600 public CPP-eligible firms. The authors can identify application status for 537. After excluding the eight largest initial CPP recipients, the main sample contains 529 firms. Of these, 424 apply, 337 are approved, and 286 ultimately receive CPP funds.

Economic message: The figure clarifies the extensive-margin structure of the paper. The authors can analyze both firm demand for aid and government allocation of aid. Most eligible firms apply, so the more interesting margin is approval conditional on application.

5.2 Table 1: Summary statistics

Read: Table 1 reports political connections, CPP outcomes, financial controls, and performance variables. About 23.4% of firms have a House Financial Services subcommittee connection, 29.9% have a connected board member, 6.8% conduct relevant lobbying, and 5.1% make relevant PAC contributions.

Economic message: The political measures are not all measuring the same thing. Correlations among political variables are positive but not so high that one variable dominates the others. This motivates the aggregate political connections index.

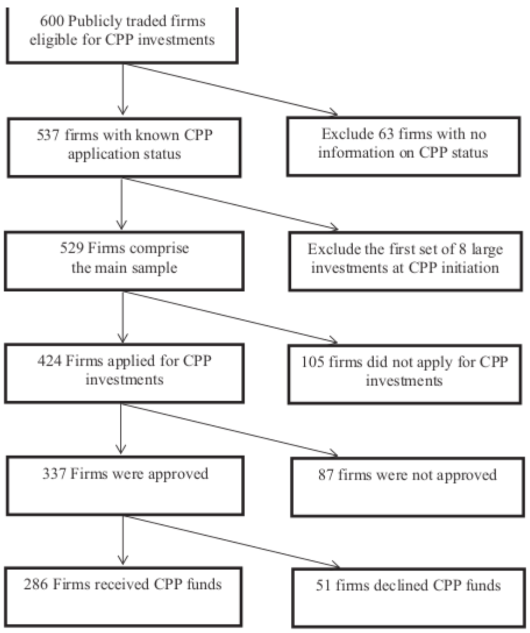

Fig. 1. Sample firms and their CPP applications. This figure illustrates the partitioning of firms eligible for CPP assistance based on the status of their CPP applications.

Table 2
Univariate evidence.
This table presents difference-in-means estimates of the likelihood of applying for CPP funds (Panel A) and the likelihood of being approved for CPP funds, conditional on applying (Panel B). For each measure of political connections, we divide our sample into firms that are connected (*Yes*) and unconnected (*No*) according to that measure. For the Political connections index, we divide our sample into quartiles and compare between the bottom quartile (*No*) and the top quartile (*Yes*). The sample for Panel A includes 529 CPP-eligible public firms with known application status. The sample for Panel B includes 424 public firms that submitted CPP applications. Variable definitions are provided in Appendix C.

Sort variable	No		Yes		Yes minus No	*t*-Statistic
	Mean	N_obs	Mean	N_obs		
***Panel A**: Application*						
House Financial Services Subcommittee	0.795	388	0.823	141	0.028	0.671
Connected board member	0.795	348	0.816	181	0.021	0.561
Lobbying	0.793	483	0.842	46	0.050	1.797
Contributions to Financial Services Subcommittee members	0.801	493	0.815	36	0.014	0.178
Political connections index	0.783	136	0.842	132	0.059	1.414
***Panel B**: Approval*						
House Financial Services Subcommittee	0.776	318	0.853	106	0.077	2.670
Connected board member	0.766	287	0.860	137	0.094	2.221
Lobbying	0.793	383	0.917	41	0.124	2.345
Contributions to Financial Services Subcommittee members	0.786	394	0.955	30	0.168	1.999
Political connections index	0.741	110	0.875	107	0.134	2.840

5.3 Table 2: Univariate evidence

Read: Panel A shows little difference in application rates between connected and unconnected firms. Panel B shows a clear approval difference: the approval rate is 85.3% for firms connected through House subcommittees versus 77.6% for others; 86.0% for firms with connected board members versus 76.6%; 91.7% for lobbying firms versus 79.3%; and 95.5% for contributor firms versus 78.6%.

Economic message: Political connections do not mainly explain who asks for help. They explain who receives help after asking. This is the first central pattern in the paper.

5.4 Table 3: Application and approval regressions

Read: Panel A estimates application regressions. The political variables are not reliably significant. Panel B estimates approval regressions among applicants. Each political measure is positive and significant in the approval specifications, including the aggregate political connections index.

Economic message: Table 3 is the main allocation result. After controlling for CAMELS-style fundamentals, size, age, industry, and regulator, politics predicts approval. The result is not a pure univariate size or distress effect.

Table 3

Application and approval for government investment funds.

This table presents estimates from logit regressions explaining the likelihood of applying for CPP funds (Panel A) and the likelihood of being approved for CPP funds, conditional on applying (Panel B). In Panel A, the dependent variable is an indicator equal to one if a firm applied to CPP. In Panel B, the dependent variable is an indicator equal to one if a firm's application to CPP was approved. All variable definitions are provided in Appendix C. The sample consists of 529 publicly traded financial firms eligible for participation in the Capital Purchase Program (CPP) with available data on program application status. The sample excludes CPP investments in the eight largest banks announced at program initiation. The standard errors (shown in brackets) are heteroskedasticity consistent. All regressions include industry and regulator fixed effects. Industries are defined by the two-digit SIC codes. Significance levels are indicated as follows: *=10%, **=5%, ***=1%.

Model number	(1)	(2)	(3)	(4)	(5)	(6)
***Panel A**: Application*						
House Financial Services Subcommittee	−0.135				−0.137	
	[0.436]				[0.447]	
Number of connected board members		0.453			0.357*	
		[0.393]			[0.209]	
Lobbying amount, scaled by size			0.047		0.045	
			[0.049]		[0.041]	
Contributions amount, scaled by size				0.184	0.015	
				[0.199]	[0.097]	
Political connections index						2.941
						[1.995]
Capital adequacy	−0.026**	−0.027*	−0.025	−0.036**	−0.024	−0.028**
	[0.013]	[0.014]	[0.015]	[0.014]	[0.016]	[0.014]
Asset quality	1.147**	1.398**	3.606**	1.621**	3.591**	1.429**
	[0.582]	[0.617]	[1.723]	[0.691]	[1.721]	[0.633]
Management quality	0.070	0.046	−0.015	0.039	−0.003	0.035
	[0.315]	[0.321]	[0.321]	[0.313]	[0.324]	[0.316]
Earnings	−0.091	−0.082	−0.080	−0.083	−0.091	−0.081
	[0.092]	[0.085]	[0.090]	[0.089]	[0.088]	[0.087]
Liquidity	−0.029	−0.038	−0.069**	−0.046	−0.076**	−0.033
	[0.028]	[0.029]	[0.033]	[0.029]	[0.032]	[0.028]
Sensitivity to market risk	0.044***	0.044***	0.045***	0.043***	0.045***	0.044***
	[0.010]	[0.010]	[0.011]	[0.010]	[0.011]	[0.010]
Foreclosures	0.007**	0.006*	0.006*	0.006**	0.006*	0.006*
	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]
Leverage	12.445***	12.643***	13.671***	12.329***	14.038***	12.207***
	[3.033]	[3.039]	[3.073]	[3.003]	[3.109]	[2.999]
Deposit-to-asset ratio	−2.222	−2.599	−1.921	−2.240	−2.123	−2.225
	[1.560]	[1.583]	[1.695]	[1.547]	[1.720]	[1.558]
Market capitalization	0.560***	0.612***	0.745***	0.586***	0.769***	0.605***
	[0.140]	[0.150]	[0.169]	[0.143]	[0.169]	[0.145]
Size	0.775***	0.690***	0.846***	0.703***	0.783***	0.692***
	[0.185]	[0.195]	[0.209]	[0.189]	[0.216]	[0.190]
Age	−0.009***	−0.009***	−0.006*	−0.008***	−0.006**	−0.008***
	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]	[0.003]
Industry fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Regulator fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R^2	0.346	0.352	0.348	0.346	0.354	0.347
N_obs	529	529	529	529	529	529
***Panel B**: Approval*						
House Financial Services Subcommittee	0.988**				1.009**	
	[0.447]				[0.462]	
Number of connected board members		0.392**			0.162**	
		[0.173]			[0.071]	
Lobbying amount, scaled by size			0.026**		0.024**	
			[0.012]		[0.010]	
Contributions amount, scaled by size				0.302**	0.136*	
				[0.134]	[0.076]	
Political connections index						4.329***
						[1.400]
Capital adequacy	0.001	−0.008	−0.024	−0.015	−0.039	−0.006
	[0.017]	[0.015]	[0.021]	[0.016]	[0.026]	[0.015]
Asset quality	0.302	0.337	0.710	0.547	0.783	0.475
	[0.410]	[0.424]	[0.723]	[0.566]	[0.744]	[0.470]
Management quality	0.282	0.316	0.313	0.279	0.302	0.308
	[0.322]	[0.322]	[0.329]	[0.320]	[0.332]	[0.322]
Earnings	0.254***	0.268***	0.271***	0.250***	0.274***	0.273***
	[0.076]	[0.080]	[0.083]	[0.080]	[0.084]	[0.081]
Liquidity	−0.012	−0.014	−0.027	−0.030	−0.028	−0.018
	[0.029]	[0.029]	[0.032]	[0.034]	[0.035]	[0.031]

Table 4 (*continued*)

Description	Different controls	Exclude top 25% capital adequacy	Exclude bottom 25% capital adequacy	Exclude New York	House Financial Services Sub-committee indicator	Unadjusted lobbying amounts	Unadjusted contributions amounts
Foreclosures	0.000	−0.008**	−0.002	−0.002	−0.002	−0.004**	−0.002*
	[0.001]	[0.003]	[0.001]	[0.001]	[0.001]	[0.002]	[0.001]
Leverage	−0.427	3.293	−2.416	−0.440	−0.848	−1.255	−1.436
	[2.806]	[4.352]	[3.332]	[2.822]	[2.645]	[2.914]	[2.887]
Deposit-to-asset ratio	−0.761	−1.196	0.231	−0.238	−0.501	−0.331	−0.937
	[1.728]	[2.636]	[1.890]	[1.814]	[1.691]	[1.796]	[1.753]
Market capitalization	0.537***	0.831***	0.389***	0.526***	0.537***	0.479***	0.694***
	[0.133]	[0.230]	[0.145]	[0.133]	[0.118]	[0.147]	[0.151]
Size	−0.268*	−0.645**	−0.263	−0.385**	−0.222	−0.268	−0.537***
	[0.158]	[0.288]	[0.175]	[0.157]	[0.137]	[0.178]	[0.192]
Age	0.009***	0.008*	0.009**	0.009**	0.009**	0.009***	0.009***
	[0.003]	[0.004]	[0.004]	[0.004]	[0.003]	[0.004]	[0.003]
Industry fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regulator fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R^2	0.176	0.261	0.148	0.174	0.166	0.186	0.186
N_obs	424	318	318	392	424	424	424

Model number	(1)	(2)	(3)	(4)	(5)	(6)
***Panel B**: All qualifying institutions*						
House Financial Services Subcommittee	0.373**				0.365**	
	[0.173]				[0.172]	
Number of connected board members		0.437***			0.256*	
		[0.130]			[0.141]	
Lobbying amount, scaled by size			0.019***		0.016***	
			[0.005]		[0.006]	
Contributions amount, scaled by size				0.206***	0.168**	
				[0.073]	[0.077]	
Political connections index						3.634***
						[1.112]
Capital adequacy	−0.017*	−0.023**	−0.044***	−0.027***	−0.050***	−0.021**
	[0.009]	[0.009]	[0.014]	[0.011]	[0.015]	[0.009]
Asset quality	0.717*	0.817*	1.635***	0.944**	1.714***	0.899**
	[0.432]	[0.440]	[0.625]	[0.448]	[0.631]	[0.438]
Management quality	0.152	0.164	0.155	0.146	0.137	0.137
	[0.218]	[0.220]	[0.222]	[0.217]	[0.223]	[0.218]
Earnings	0.140**	0.158**	0.192***	0.142**	0.197***	0.158**
	[0.059]	[0.062]	[0.064]	[0.061]	[0.065]	[0.062]
Liquidity	−0.019	−0.026	−0.038	−0.037*	−0.042*	−0.026
	[0.020]	[0.021]	[0.024]	[0.022]	[0.024]	[0.020]
Sensitivity to market risk	−0.020**	−0.019**	−0.022***	−0.019**	−0.021**	−0.019**
	[0.008]	[0.008]	[0.008]	[0.008]	[0.008]	[0.008]
Foreclosures	−0.001	0.000	−0.002*	0.000	−0.002	0.000
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
Leverage	4.441**	4.674**	4.304**	5.003**	4.114**	4.421**
	[1.928]	[1.920]	[2.017]	[1.948]	[2.025]	[1.923]
Deposit-to-asset ratio	−0.738	−1.085	−1.275	−1.244	−1.315	−0.814
	[1.097]	[1.095]	[1.176]	[1.108]	[1.196]	[1.097]
Market capitalization	0.005	0.006	−0.005	0.032	−0.016	−0.019
	[0.088]	[0.091]	[0.102]	[0.092]	[0.102]	[0.089]
Size	0.230**	0.113	0.131	0.101	0.048	0.125
	[0.106]	[0.115]	[0.125]	[0.119]	[0.132]	[0.112]
Age	0.000	0.000	0.001	0.001	0.001	0.000
	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]
Industry fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Regulator fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R^2	0.110	0.124	0.145	0.120	0.152	0.121
N_obs	600	600	600	600	600	600
***Panel C**: Placebo tests*						
House Financial Services Subcommittee	0.011				0.038	

5.5 Table 4: Robustness and placebo tests

Read: Table 4 changes controls, samples, and political measures. It uses alternative CAMELS proxies, removes high- and low-capital firms, excludes New York-headquartered firms, uses unscaled lobbying and contribution amounts, expands to all qualifying institutions, and then replaces true political variables with placebo variables.

Economic message: True contemporaneous political connections remain predictive of approval; placebo connections generally do not. This makes it harder to argue that the main result is just a generic effect of firm prominence, old political habits, or irrelevant lobbying.

5.6 Table 5: Systemic importance

Read: The table controls for systemic importance by excluding the largest firms, adding nonlinear size terms, adding a Peer Group 1 indicator, using size-matched and state-size-matched samples, and including DCoVaR.

Economic message: The political connections index remains positive and significant. This directly addresses the concern that connected firms were approved because they were too big or too systemically important to fail.

Table 5
Systemic importance.

This table presents estimates from logit regressions explaining the likelihood of CPP application approval. The dependent variable is an indicator equal to one if a firm's application for CPP was approved. In column 3, systemic importance is indicated by the Peer Group 1 indicator, which equals one for the set of firms with book assets of at least $3 billion. In columns 4 and 5, we consider subsamples of all banks that are connected according to the political connections index and their best-match unconnected counterpart. In column 4, banks are matched by their size (book assets). In column 5, banks are matched by size to banks headquartered in the same state. In both columns 4 and 5, we require that the size difference between matched pairs not exceed 10%. In column 6, systemic importance is measured by ΔCoVaR, a measure suggested by Adrian and Brunnermeier (2011), and defined as the value-at-risk of the financial system conditional on an institution being under distress. Variable definitions are shown in Appendix C. The sample consists of 529 publicly traded financial firms eligible for participation in the Capital Purchase Program (CPP) with available data on program application status. The sample excludes CPP investments in the eight largest banks announced at program initiation. All regressions include industry and regulator fixed effects. The standard errors (in brackets) are heteroskedasticity consistent. Significance levels are indicated as follows: *=10%, **=5%, ***=1%.

Description	Exclude top 5% size	Higher-order powers of size	Peer Group 1	Size-matched sample	State-size-matched sample	ΔCoVaR
Model number	(1)	(2)	(3)	(4)	(5)	(6)
Political connections index	3.990**	3.247**	4.545***	4.743**	12.912**	4.792**
	[1.792]	[1.576]	[1.764]	[2.016]	[5.147]	[1.963]
Capital adequacy	−0.118**	−0.066**	−0.017	−0.148***	−0.552***	−0.013
	[0.047]	[0.028]	[0.015]	[0.057]	[0.195]	[0.020]
Asset quality	0.499	0.609	0.622	1.181	1.753	0.500
	[0.607]	[0.602]	[0.620]	[1.011]	[2.702]	[0.639]
Management quality	0.344	0.360	0.337	0.074	0.574	0.069
	[0.322]	[0.323]	[0.325]	[0.353]	[0.870]	[0.355]
Earnings	0.240***	0.296***	0.332***	0.475***	−0.204	0.317***
	[0.080]	[0.086]	[0.084]	[0.106]	[0.367]	[0.099]
Liquidity	−0.028	−0.055	−0.007	−0.036	0.060	−0.080**
	[0.038]	[0.038]	[0.035]	[0.062]	[0.145]	[0.033]
Sensitivity to market risk	0.008	0.003	0.007	−0.003	0.038	0.001
	[0.015]	[0.015]	[0.015]	[0.015]	[0.032]	[0.015]
Foreclosures	−0.002**	−0.003**	−0.001	−0.005**	−0.024*	−0.006***
	[0.001]	[0.001]	[0.001]	[0.002]	[0.012]	[0.002]
Leverage	−2.659	−2.277	−1.590	−3.296	−5.314	2.682
	[2.840]	[2.869]	[2.848]	[3.872]	[9.874]	[3.428]
Deposit-to-asset ratio	−0.709	−0.006	−0.388	−0.274	11.532*	0.961
	[1.733]	[1.763]	[1.745]	[2.093]	[6.957]	[1.786]
Market capitalization	0.715***	0.526***	0.410***	0.454***	0.431	0.422**
	[0.158]	[0.159]	[0.133]	[0.170]	[0.671]	[0.169]
Size	−0.635***	-4.347	−0.545***	−0.233	−0.552	0.026
	[0.197]	[12.015]	[0.169]	[0.221]	[1.065]	[0.213]
Age	0.008**	0.009***	0.008**	0.003	0.024**	0.004
	[0.003]	[0.003]	[0.003]	[0.004]	[0.010]	[0.004]
Size^2		0.032				
		[0.842]				
Size^3		0.005				
		[0.020]				
Peer Group 1 indicator			3.456***			
			[1.156]			
ΔCoVaR						1.780*
						[0.938]
Industry fixed effects?	Yes	0.058	0.094	Yes	Yes	Yes
Regulator fixed effects?	Yes	318	392	Yes	Yes	Yes
Pseudo-R^2	0.178	0.200	0.217	0.194	0.361	0.201
N_obs	403	424	424	362	84	363

Table 6
Investment performance.

This table presents estimates from panel regressions where the dependent variable is a measure of firm performance: return on assets, Tobin's *Q*, and stock returns. The sample consists of the publicly traded firms that received CPP funds. All variables are measured at quarterly frequency. Panel A considers the period before CPP investments, from the second quarter of 2006 to the last quarter of 2008. Bank-level controls include *Capital adequacy*, *Asset quality*, *Liquidity*, *Sensitivity to market risk*, *Foreclosures*, *Leverage*, *Deposit-to-asset ratio*, and *Size*. The sample period in Panels B and C covers the second quarter of 2006 through the second quarter of 2010. *After CPP investment* is an indicator variable equal to zero before 2009 and one thereafter. *ROA* is return on assets. *Tobin's Q* is the book value of assets minus the book value of equity plus the market value of equity divided by the book value of assets. In column 3, we use an alternative definition of Tobin's *Q* to control for outliers, which divides the firm's market value by 0.9 times its book value plus 0.1 times its market value. *Market-adjusted returns* are raw returns minus the returns on the CRSP value-weighted index. *Industry-adjusted returns* are raw returns minus industry returns, where industries are defined by the two-digit SIC codes. All regressions in Panels B and C include bank fixed effects. Variable definitions are provided in Appendix C. The standard errors (in brackets) are heteroskedasticity consistent and clustered at the bank level. Significance levels are indicated as follows: *=10%, **=5%, ***=1%.

Dependent variable	ROA	Tobin's *Q*	Tobin's *Q* (alternative definition)	Raw returns	Market-adjusted returns	Industry-adjusted returns
Model number	(1)	(2)	(3)	(4)	(5)	(6)
***Panel A**: Pre-CPP performance*						
Political connections index	−0.336	3.480	3.047	−0.004	−0.038	−0.037
	[0.454]	[3.223]	[2.819]	[0.025]	[0.024]	[0.025]
Bank-level controls?	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.151	0.097	0.097	0.015	0.004	0.008
N_obs	3,945	3,628	3,628	3,488	3,488	3,488
***Panel B**: Univariate tests*						
After CPP investment	−0.790*	−4.080***	−3.683***	−0.017	−0.086***	−0.107***
	[0.416]	[0.803]	[0.713]	[0.028]	[0.028]	[0.028]
After CPP investment × Political connections index	−1.439**	−3.254**	−2.849**	−0.091**	−0.106**	−0.083**
	[0.722]	[1.510]	[1.339]	[0.041]	[0.052]	[0.039]
Bank fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.468	0.648	0.648	0.042	0.044	0.063
N_obs	5,906	5,438	5,438	5,254	5,254	5,254
***Panel C**: Multivariate tests*						
After CPP investment	−0.391	−1.265	−1.187	−0.010	−0.108***	−0.124***
	[0.400]	[1.014]	[0.901]	[0.030]	[0.029]	[0.028]
After CPP investment × Political connections index	−1.604**	−4.967**	−4.364**	−0.077**	−0.111**	−0.088*
	[0.688]	[1.961]	[1.742]	[0.035]	[0.053]	[0.051]
Capital adequacy	0.002	0.018*	0.016*	0.001	0.000	0.001
	[0.004]	[0.010]	[0.009]	[0.001]	[0.000]	[0.001]
Asset quality	0.173	0.562	0.505	0.006	0.012	0.007
	[0.236]	[0.352]	[0.314]	[0.010]	[0.007]	[0.007]
Liquidity	0.029***	0.042	0.034	0.001	0.003**	0.004***
	[0.010]	[0.053]	[0.045]	[0.001]	[0.001]	[0.001]
Sensitivity to market risk	−0.001	−0.001	−0.001	0.000	0.001	0.001
	[0.007]	[0.021]	[0.019]	[0.001]	[0.001]	[0.001]
Foreclosures	−0.003***	−0.003**	−0.003**	−0.000***	0.000	0.000
	[0.000]	[0.001]	[0.001]	[0.000]	[0.000]	[0.000]
Leverage	−0.376	−17.796***	−15.805***	−0.358***	−0.056	−0.109
	[0.740]	[2.586]	[2.298]	[0.072]	[0.077]	[0.071]
Deposit-to-asset ratio	1.064	18.474***	16.408***	0.377***	0.121	0.187**
	[0.743]	[3.031]	[2.689]	[0.086]	[0.090]	[0.083]
Size	−0.311	−10.106***	−8.997***	−0.136***	0.047*	0.019
	[0.270]	[1.127]	[1.004]	[0.019]	[0.026]	[0.024]
Bank fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.461	0.731	0.731	0.057	0.045	0.061
N_obs	5,906	5,438	5,438	5,254	5,254	5,254

5.7 Table 6: Investment performance

Read: Panel A finds no strong pre-CPP performance difference associated with the political connections index. Panels B and C show negative coefficients on After CPP times Political Connections Index across ROA, Tobin's Q, and return outcomes, with bank fixed effects and, in Panel C, time-varying bank controls.

Economic message: This table is the mechanism test. If political connections simply helped regulators identify better firms, connected recipients should not underperform. Instead, they do, which supports the political-distortion interpretation.

6 Short summary

- The paper studies whether political connections affected the allocation of CPP/TARP capital during the 2008–2009 crisis.
- The authors hand-collect application status for public CPP-eligible financial institutions.
- Most eligible firms apply for CPP, and political connections do not strongly predict application.
- Among applicants, politically connected firms are more likely to be approved for CPP capital.
- The approval result survives financial controls, regulator and industry fixed effects, systemic-importance controls, matched samples, and placebo tests.
- Connected recipients perform worse after receiving CPP funds, according to ROA, Tobin's Q, and stock-return measures.
- The evidence supports the view that political influence distorted government capital allocation rather than improved it through better information.

7 Conclusion

Duchin and Sosyura provide evidence that political connections mattered in one of the most important government investment programs in U.S. financial history. The key empirical pattern is two-sided: connections did not matter much for the decision to apply for CPP, but they mattered for approval conditional on application.

The performance evidence is crucial. Politically connected recipients underperformed unconnected recipients after CPP, while they did not display meaningful performance advantages before the program. This makes the information-transmission interpretation less persuasive and makes the political-distortion interpretation more persuasive.

The broader lesson is that emergency government investment programs can be shaped by political networks even when the program has formal criteria and public oversight. The policy challenge is to design crisis interventions that can move quickly while preserving allocative discipline and limiting political favoritism.